

LAPLACE TRANSFORM

Def. Exponential order

A function $f(t)$ is said to be of exponential order if

$$\lim_{t \rightarrow \infty} e^{-st} f(t) = 0$$

Example 1 Show that x^n is of exponential order as $x \rightarrow \infty$, $n > 0$.

Solution :

$$\begin{aligned} \lim_{x \rightarrow \infty} e^{-ax} x^n &= \lim_{x \rightarrow \infty} \frac{x^n}{e^{ax}} \left[\frac{\infty}{\infty} \text{ i.e., Indeterminant form} \right] \\ &= \lim_{x \rightarrow \infty} \frac{n x^{n-1}}{a e^{ax}} \left[\frac{\infty}{\infty} \text{ i.e., Indeterminant form} \right] \\ &\quad \text{[Apply L' Hospital Rule]} \\ &= \lim_{x \rightarrow \infty} \frac{n(n-1) \dots 1}{a^n e^{ax}} \text{ [Repeating this process we get]} \\ &= \lim_{x \rightarrow \infty} \frac{n!}{a^n e^{ax}} \text{ [Applying L'Hospital's rule]} \\ &= \frac{n!}{\infty} = 0 \end{aligned}$$

Hence x^n is of exponential order.

Example Show that t^2 is of exponential order.

$$\begin{aligned} \text{Solution : } \lim_{t \rightarrow \infty} e^{-st} t^2 &= \lim_{t \rightarrow \infty} \frac{t^2}{e^{st}} \left[\frac{\infty}{\infty} \text{ i.e., Indeterminant form} \right] \\ &\quad \text{[Apply L'Hospital's rule]} \\ &= \lim_{t \rightarrow \infty} \frac{2t}{s e^{st}} \left[\frac{\infty}{\infty} \text{ form} \right] \text{ [Apply L'Hospital's Rule]} \\ &= \lim_{t \rightarrow \infty} \frac{2}{s^2 e^{st}} = \frac{2}{\infty} \\ &= 0 \end{aligned}$$

Hence t^2 is of exponential order.

Example Show that the function

$f(t) = e^{t^2}$ is not of exponential order.

$$\begin{aligned} \text{Solution : } \lim_{t \rightarrow \infty} e^{-st} e^{t^2} &= \lim_{t \rightarrow \infty} e^{-st + t^2} \\ &= e^{\infty} = \infty \end{aligned}$$

So $f(t) = e^{t^2}$ is not of exponential order.

Define function of class A.

Solution : A function which is sectionally continuous over any finite interval and is of exponential order is known as a function of class A.

◆ Important Result

$$(1) \quad L[1] = \frac{1}{s} \quad \text{where } s > 0$$

$$(2) \quad L[t^n] = \frac{n!}{s^{n+1}} \quad \text{where } n = 0, 1, 2, \dots$$

$$(3) \quad L[t^n] = \frac{\Gamma n+1}{s^{n+1}} \quad \text{where } n \text{ is not a integer.}$$

$$(4) \quad L[e^{at}] = \frac{1}{s-a} \quad \text{where } s > a \text{ or } s-a > 0$$

$$(5) \quad L[e^{-at}] = \frac{1}{s+a} \quad \text{where } s+a > 0$$

$$(6) \quad L[\sin at] = \frac{a}{s^2+a^2} \quad \text{where } s > 0$$

$$(7) \quad L[\cos at] = \frac{s}{s^2+a^2} \quad \text{where } s > 0$$

$$(8) \quad L[\sinh at] = \frac{a}{s^2-a^2} \quad \text{where } s > |a| \text{ or } s^2 > a^2$$

$$(9) \quad L[\cosh at] = \frac{s}{s^2-a^2} \quad \text{where } s^2 > a^2$$

$$(10) \quad L[af(t) \pm bg(t)] = a L[f(t)] \pm b L[g(t)] \quad [\text{Linearity property}]$$

Note : (1) $e^x = 1 + \frac{x}{1} + \frac{x^2}{2} + \dots$

$$e^\infty = 1 + \frac{\infty}{1} + \frac{\infty^2}{2} + \dots$$

$$(2) \quad e^{-\infty} = \frac{1}{e^\infty} = \frac{1}{\infty} = 0$$

$$(3) \quad \Gamma_{n+1} = n!$$

$$(4) \quad \Gamma_{n+1} = \int_0^{\infty} x^n e^{-x} dx$$

$$(5) \quad \Gamma_{n+1} = n \Gamma_n$$

$$(6) \quad \Gamma_{1/2} = \sqrt{\pi}$$

$$(7) \quad \int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]$$

$$(8) \quad \int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2 + b^2} [a \cos bx + b \sin bx]$$

$$(9) \quad \sin^3 \theta = \frac{1}{4} [3 \sin \theta - \sin 3 \theta]$$

$$(10) \quad \cos^3 \theta = \frac{1}{4} [\cos 3 \theta + 3 \cos \theta]$$

$$(11) \quad \sin A \cos B = \frac{1}{2} [\sin (A + B) + \sin (A - B)]$$

$$(12) \quad \cos A \sin B = \frac{1}{2} [\sin (A + B) - \sin (A - B)]$$

$$(13) \quad \cos A \cos B = \frac{1}{2} [\cos (A + B) + \cos (A - B)]$$

$$(14) \quad \sin A \sin B = -\frac{1}{2} [\cos (A + B) - \cos (A - B)]$$

5.2 TRANSFORMS OF ELEMENTARY FUNCTIONS - BASIC PROPERTIES

Result (1) : Prove that $L[1] = \frac{1}{s}$ where $s > 0$

Proof : We know that $L[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$

Here $f(t) = 1$

$$\begin{aligned} \therefore L[1] &= \int_0^{\infty} e^{-st} dt = \left[\frac{e^{-st}}{-s} \right]_0^{\infty} \\ &= -\frac{1}{s} \left[e^{-st} \right]_0^{\infty} = -\frac{1}{s} [e^{-\infty} - e^{-0}] \\ &= -\frac{1}{s} [0 - 1] \text{ by note (2)} \\ &= \frac{1}{s}, s > 0 \end{aligned}$$

Result (2) : Prove that $L[t^n] = \frac{n!}{s^{n+1}}$ [$n = 0, 1, 2, \dots$]

Proof : We know that

$$\begin{aligned} L[f(t)] &= \int_0^{\infty} e^{-st} f(t) dt \\ L[t^n] &= \int_0^{\infty} e^{-st} t^n dt = \int_0^{\infty} t^n d\left[\frac{e^{-st}}{-s}\right] \\ &= t^n \left[\frac{e^{-st}}{-s}\right]_0^{\infty} - \int_0^{\infty} \frac{e^{-st}}{-s} n t^{n-1} dt \\ &= (0 - 0) + \frac{n}{s} \int_0^{\infty} e^{-st} t^{n-1} dt \end{aligned}$$

$$\text{i.e., } L[t^n] = \frac{n}{s} L[t^{n-1}]$$

$$\text{Similarly } L[t^{n-1}] = \frac{n-1}{s} L[t^{n-2}]$$

$$L[t^{n-2}] = \frac{n-2}{s} L[t^{n-3}]$$

.....
.....

$$\begin{aligned} L[t^{n-(n-1)}] &= \frac{n-(n-1)}{s} L[t^{n-(n-1)-1}] \\ &= \frac{1}{s} L[t^0] = \frac{1}{s} L[1] = \frac{1}{s} \frac{1}{s} \end{aligned}$$

$$\begin{aligned} \therefore L[t^n] &= \frac{n}{s} \frac{n-1}{s} \dots \frac{2}{s} \frac{1}{s} \frac{1}{s} = \frac{n!}{s^n} \frac{1}{s} \\ &= \frac{n!}{s^{n+1}} \text{ where } [n = 0, 1, 2, \dots] \end{aligned}$$

Result (3) Prove that $L[t^n] = \frac{\Gamma_{n+1}}{s^{n+1}}$ where n is not a integer.

Proof : We know that

$$\begin{aligned} L[f(t)] &= \int_0^{\infty} e^{-st} f(t) dt \\ L[t^n] &= \int_0^{\infty} e^{-st} t^n dt \end{aligned}$$

$$\begin{array}{ll} \text{Put } st = x & \text{as } t \rightarrow 0 \Rightarrow x \rightarrow 0 \\ s \, dt = dx & \text{as } t \rightarrow \infty \Rightarrow x \rightarrow \infty \end{array}$$

$$\begin{aligned} &= \int_0^{\infty} e^{-x} \left(\frac{x}{s} \right)^n \frac{dx}{s} \\ &= \int_0^{\infty} e^{-x} \frac{x^n}{s^{n+1}} dx \\ &= \frac{1}{s^{n+1}} \int_0^{\infty} x^n e^{-x} dx \end{aligned}$$

$$\text{i.e., } L[t^n] = \frac{\Gamma_{n+1}}{s^{n+1}} \quad \left[\because \int_0^{\infty} x^n e^{-x} dx = \Gamma_{n+1} \right]$$

when n is a positive integer.

we get $\Gamma_{n+1} = n!$

$$L[t^n] = \frac{n!}{s^{n+1}}$$

II. PROBLEMS BASED ON TRANSFORMS OF ELEMENTARY FUNCTIONS - BASIC PROPERTIES

Example 1 Find $L[t]$

Solution : $L[t^n] = \frac{n!}{s^{n+1}}$ [we know that]

$$L[t] = \frac{1!}{s^{1+1}} = \frac{1}{s^2}$$

Example 2 Find $L[t^3]$

Solution : We know that $L[t^n] = \frac{n!}{s^{n+1}}$

$$L[t^3] = \frac{3!}{s^{3+1}} = \frac{6}{s^4}$$

Example 3 Find $L[\sqrt{t}]$

Solution : We know that $L[t^n] = \frac{\Gamma_{n+1}}{s^{n+1}}$

$$L[\sqrt{t}] = L[t^{1/2}] = \frac{\Gamma_{1/2+1}}{s^{1/2+1}}$$

$$\begin{aligned}
&= \frac{\frac{1}{2} \Gamma_{1/2}}{s^{3/2}} \quad [\because \Gamma_{n+1} = n \Gamma_n ; \Gamma_{1/2} = \sqrt{\pi}] \\
&= \frac{\Gamma_{1/2}}{2 s^{3/2}} = \frac{\sqrt{\pi}}{2 s^{3/2}}
\end{aligned}$$

Example 4. Find $L[t^{3/2}]$

Solution :

We know that $L[t^n] = \frac{\Gamma_{n+1}}{s^{n+1}}$

$$\begin{aligned}
L[t^{3/2}] &= \frac{\Gamma_{3/2+1}}{s^{3/2+1}} = \frac{\frac{3}{2} \Gamma_{3/2}}{s^{5/2}} \\
&= \frac{\frac{3}{2} \Gamma_{1/2+1}}{s^{5/2}} = \frac{\left(\frac{3}{2}\right) \left(\frac{1}{2}\right) \Gamma_{1/2}}{s^{5/2}} \\
&= \frac{\left(\frac{3}{4}\right) \sqrt{\pi}}{s^{5/2}} \quad [\because \Gamma_{1/2} = \sqrt{\pi}] \\
&= \frac{3 \sqrt{\pi}}{4 s^{5/2}}
\end{aligned}$$

Example 5.2.5. Find $L\left[\frac{1}{\sqrt{t}}\right]$

Solution : We know that $L[t^n] = \frac{\Gamma_{n+1}}{s^{n+1}}$

$$\begin{aligned}
L\left[\frac{1}{\sqrt{t}}\right] &= L[t^{-1/2}] = \frac{\Gamma_{-1/2+1}}{s^{-1/2+1}} \\
&= \frac{\Gamma_{1/2}}{s^{1/2}} \\
&= \frac{\sqrt{\pi}}{\sqrt{s}} = \sqrt{\frac{\pi}{s}} \quad [\because \Gamma_{1/2} = \sqrt{\pi}]
\end{aligned}$$

Result 4. Prove that $L[e^{at}] = \frac{1}{s-a}$ where $s > a$.

Proof : We know that

$$L[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$$

$$\begin{aligned}
 L[e^{at}] &= \int_0^{\infty} e^{-st} e^{at} dt = \int_0^{\infty} e^{-(s-a)t} dt \\
 &= \left[\frac{e^{-(s-a)t}}{-(s-a)} \right]_0^{\infty} = -\frac{1}{s-a} \left[e^{-(s-a)t} \right]_0^{\infty} \\
 &= \frac{-1}{s-a} [0 - 1] = \frac{1}{s-a} \text{ where } s-a > 0
 \end{aligned}$$

Example 6. Find the value $L[e^{3t}]$

Solution : We know that

$$L[e^{at}] = \frac{1}{s-a}$$

$$L[e^{3t}] = \frac{1}{s-3}$$

Example 7 Find $L[e^{3t+5}]$

Solution :

W.K.T $L[e^{at}] = \frac{1}{s-a}$

$$\begin{aligned}
 L[e^{3t+5}] &= L[e^{3t} e^5] \\
 &= e^5 L[e^{3t}] = e^5 \left[\frac{1}{s-3} \right] = \frac{e^5}{s-3}
 \end{aligned}$$

Example 8 Find $L\left[\frac{e^{at}}{a}\right]$

Solution : W.K.T $L[e^{at}] = \frac{1}{s-a}$

$$L\left[\frac{e^{at}}{a}\right] = \frac{1}{a} L[e^{at}] = \frac{1}{a} \left[\frac{1}{s-a} \right]$$

Example 9 Find $L[2^t]$

W.K.T. $L[e^{at}] = \frac{1}{s-a}$

$$\begin{aligned}
 L[2^t] &= L\left[e^{\log 2^t}\right] \\
 &= L\left[e^{t \log 2}\right] \\
 &= L\left[e^{(\log 2)t}\right] \\
 &= \frac{1}{s - \log 2}
 \end{aligned}$$

Result 5. Prove that $L[e^{-at}] = \frac{1}{s+a}$, $(s+a) > 0$

Proof : W.K.T. $L[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$

$$\begin{aligned}
 L[e^{-at}] &= \int_0^{\infty} e^{-st} e^{-at} dt \\
 &= \int_0^{\infty} e^{-(s+a)t} dt \\
 &= \left[\frac{e^{-(s+a)t}}{-(s+a)} \right]_0^{\infty} = -\frac{1}{s+a} \left[e^{-(s+a)t} \right]_0^{\infty} \\
 &= -\frac{1}{s+a} [0 - 1] \\
 &= \frac{1}{s+a} \text{ where } (s+a) > 0
 \end{aligned}$$

Example 10. Find $L[e^{-bt}]$

Solution : W.K.T $L[e^{-at}] = \frac{1}{s+a}$

$$L[e^{-bt}] = \frac{1}{s+b}$$

Example 11. Find $L[2e^{-3t}]$

Solution : W.K.T. $L[e^{-at}] = \frac{1}{s+a}$

$$\begin{aligned}
 L[2e^{-3t}] &= 2L[e^{-3t}] \\
 &= 2 \left[\frac{1}{s+3} \right] = \left[\frac{2}{s+3} \right]
 \end{aligned}$$

Result 6. Prove that $L[\sin at] = \frac{a}{s^2 + a^2}$ ($s > 0$)

Proof : W.K.T. $L[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$

$$L[\sin at] = \int_0^{\infty} e^{-st} \sin at dt$$

$$= \left[\frac{e^{-st}}{s^2 + a^2} [-s \sin at - a \cos at] \right]_0^{\infty} \text{ by Note 7.}$$

$$\int e^{ax} b(x) dx$$

$$= \frac{e^{ax}}{a^2 + b^2} [a \sin bx - b \cos bx]$$

$$= 0 - \left[\frac{(-a)}{s^2 + a^2} \right] = \frac{a}{s^2 + a^2} \text{ where } s > 0.$$

Example 5.2.12. Find $L[\sin 2t]$

Solution : W.K.T $L[\sin at] = \frac{a}{s^2 + a^2}$

$$\begin{aligned} L[\sin 2t] &= \frac{2}{s^2 + 2^2} \\ &= \frac{2}{s^2 + 4} \end{aligned}$$

Example 5.2.13. Find $L[\sin \pi t]$

Solution : W.K.T $L[\sin at] = \frac{a}{s^2 + a^2}$

$$L[\sin \pi t] = \frac{\pi}{s^2 + \pi^2}$$

Result : 7. Prove that $L[\cos at] = \frac{s}{s^2 + a^2} \ (s > 0)$

Proof : W.K.T. $L[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$

$$\begin{aligned} L[\cos at] &= \int_0^{\infty} e^{-st} \cos at \, dt \\ &= \left[\frac{e^{-st}}{s^2 + a^2} [-s \cos at + a \sin at] \right]_0^{\infty} \\ &= 0 - \left[\frac{1}{s^2 + a^2} (-s) \right] \\ &= \frac{s}{s^2 + a^2} \ (s > 0) \end{aligned}$$

Example 5.2.14. Find $L[\cos 2t]$

Solution : W.K.T. $L[\cos at] = \frac{s}{s^2 + a^2}$

$$L[\cos 2t] = \frac{s}{s^2 + 4}$$

Example 15 Prove that $L[\cos at] = \frac{s}{s^2 + a^2}$ and $L[\sin at] = \frac{a}{s^2 + a^2}$ Mathematics II

Solution : By Euler's theorem

$$e^{ix} = \cos x + i \sin x$$

$$e^{iat} = \cos at + i \sin at$$

$$\begin{aligned} L[e^{iat}] &= L[\cos at + i \sin at] \\ &= L[\cos at] + i L[\sin at] \end{aligned}$$

$$\begin{aligned} L[\cos at] + i L[\sin at] &= L[e^{iat}] \\ &= \frac{1}{s - ia} \\ &= \left[\frac{1}{s - ia} \right] \left[\frac{s + ia}{s + ia} \right] \\ &= \frac{s + ia}{s^2 + a^2} \end{aligned}$$

Equating real & Imaginary parts we get

$$L[\cos at] = \frac{s}{s^2 + a^2}$$

$$L[\sin at] = \frac{a}{s^2 + a^2}$$

Example 16 Find $L[\cos (at + b)]$

Solution : $L[\cos (at + b)]$

$$\begin{aligned} &= L[\cos at \cos b - \sin at \sin b] \\ &= \cos b L[\cos at] - \sin b L[\sin at] \\ &= \cos b \left[\frac{s}{s^2 + a^2} \right] - \sin b \left[\frac{a}{s^2 + a^2} \right] \\ &= \frac{s \cos b - a \sin b}{s^2 + a^2} \end{aligned}$$

Example 17 Find $L[\sin^2 2t]$

$$\begin{aligned} \text{Solution : } L[\sin^2 2t] &= L\left[\frac{1 - \cos 4t}{2}\right] = \frac{1}{2} L[1 - \cos 4t] \\ &= \frac{1}{2} [L[1] - L[\cos 4t]] \\ &= \frac{1}{2} \left[\frac{1}{s} - \frac{s}{s^2 + 16} \right] \end{aligned}$$

Solution : $L[\sin 5t \cos 2t] = \frac{1}{2} L[\sin 7t + \sin 3t]$ by Note 11.

$$= \frac{1}{2} \left[\frac{7}{s^2 + 49} + \frac{3}{s^2 + 9} \right]$$

Example 19 Find $L[(\sin t - \cos t)^2]$

Solution : $L[(\sin t - \cos t)^2] = L[\sin^2 t + \cos^2 t - 2 \sin t \cos t]$

$$= L[1 - \sin 2t] = L[1] - L[\sin 2t]$$
$$= \frac{1}{s} - \frac{2}{s^2 + 4}$$

Result 8. Prove that $L[\sinh at] = \frac{a}{s^2 - a^2}$ where $s > |a|$

Proof : $\sinh at = \frac{e^{at} - e^{-at}}{2}$

$$\begin{aligned} L[\sinh at] &= L \left[\frac{e^{at} - e^{-at}}{2} \right] \\ &= \frac{1}{2} L[e^{at} - e^{-at}] = \frac{1}{2} [L[e^{at}] - L[e^{-at}]] \\ &= \frac{1}{2} \left[\frac{1}{s-a} - \frac{1}{s+a} \right] = \frac{1}{2} \left[\frac{s+a - s+a}{s^2 - a^2} \right] \\ &= \frac{1}{2} \left[\frac{2a}{s^2 - a^2} \right] = \frac{a}{s^2 - a^2}, \quad s > |a| \end{aligned}$$

Result 9. Prove that $L[\cosh at] = \frac{s}{s^2 - a^2}$, $s > |a|$

Proof : $\cosh at = \frac{e^{at} + e^{-at}}{2}$

$$\begin{aligned} L[\cosh at] &= L \left[\frac{e^{at} + e^{-at}}{2} \right] \\ &= \frac{1}{2} L[e^{at} + e^{-at}] = \frac{1}{2} [L[e^{at}] + L[e^{-at}]] \\ &= \frac{1}{2} \left[\frac{1}{s-a} + \frac{1}{s+a} \right] = \frac{1}{2} \left[\frac{s+a + s-a}{s^2 - a^2} \right] \\ &= \frac{1}{2} \left[\frac{2s}{s^2 - a^2} \right] = \frac{s}{s^2 - a^2}, \quad s > |a| \end{aligned}$$

Result 10. Linearity property.

Prove that $L [a f(t) \pm b g(t)] = a L [f(t)] \pm b L [g(t)]$

Proof : W.K.T. $L [f(t)] = \int_0^{\infty} e^{-st} f(t) dt$

$$\begin{aligned} L[af(t) \pm bg(t)] &= \int_0^{\infty} e^{-st} [af(t) \pm bg(t)] dt \\ &= \int_0^{\infty} e^{-st} af(t) dt \pm \int_0^{\infty} e^{-st} bg(t) dt \\ &= a \int_0^{\infty} e^{-st} f(t) dt \pm b \int_0^{\infty} e^{-st} g(t) dt \\ &= a L[f(t)] \pm b L[g(t)] \end{aligned}$$

Example $L [e^{4t} + t^4 + 7]$

Solution : $L[e^{4t} + t^4 + 7]$

$$\begin{aligned} &= L [e^{4t}] + L [t^4] + L [7] \\ &= \frac{1}{s-4} + \frac{4!}{s^5} + 7 L [1] \\ &= \frac{1}{s-4} + \frac{24}{s^5} + 7 \left[\frac{1}{s} \right] \end{aligned}$$

Example 5.2.26. Find $L [f(t)]$ if $f(t) = \begin{cases} e^{-t}, & 0 < t < 4 \\ 0, & t > 4 \end{cases}$

Solution : W.K.T. $L[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$

$$\begin{aligned} &= \int_0^4 e^{-st} e^{-t} dt + \int_4^{\infty} e^{-st} 0 dt \\ &= \int_0^4 e^{-(s+1)t} dt + 0 \\ &= \left[\frac{e^{-(s+1)t}}{-(s+1)} \right]_0^4 = \frac{-1}{(s+1)} \left[e^{-(s+1)t} \right]_0^4 \\ &= \frac{-1}{s+1} \left[e^{-4(s+1)} - 1 \right] = \frac{1}{s+1} [1 - e^{-4(s+1)}] \end{aligned}$$

Result 11. Prove that $L[f'(t)] = s L[f(t)] - f(0)$

Proof : W.K.T. $L[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$

$$L[f'(t)] = \int_0^{\infty} e^{-st} f'(t) dt$$

$$= \int_0^{\infty} e^{-st} d[f(t)]$$

$$= \left[e^{-st} f(t) \right]_0^{\infty} - \int_0^{\infty} f(t) (-s) e^{-st} dt$$

$$= [0 - f(0)] + s \int_0^{\infty} e^{-st} f(t) dt$$

$$= -f(0) + s L[f(t)]$$

$$= s L[f(t)] - f(0)$$

Result 12. Prove that $L[f''(t)] = s^2 L[f(t)] - s f(0) - f'(0)$

Proof : W.K.T. $L[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$

$$L[f''(t)] = \int_0^{\infty} e^{-st} f''(t) dt$$

$$= \int_0^{\infty} e^{-st} d[f'(t)]$$

$$= \left[e^{-st} f'(t) \right]_0^{\infty} - \int_0^{\infty} f'(t) (-s) e^{-st} dt$$

$$\begin{aligned}
&= [0 - f'(0)] + s \int_0^{\infty} e^{-st} f'(t) dt \\
&= -f'(0) + s L[f'(t)] \\
&= -f'(0) + s [sL[f(t)] - f(0)] \text{ by result (1.1)} \\
&= s^2 L[f(t)] - sf(0) - f'(0)
\end{aligned}$$

Note : (15)

$$L[f^n(t)] = s^n L[f(t)] - s^{n-1}f(0) - s^{n-2}f'(0) - \dots - f^{n-1}(0)$$

Result : 13. FIRST SHIFTING THEOREM

- If $L[f(t)] = \varphi(s)$ then $L[e^{at}f(t)] = \varphi(s-a)$
- If $L[f(t)] = \varphi(s)$ then $L[e^{-at}f(t)] = \varphi(s+a)$

Proof : W.K.T $L[f(t)] = \varphi(s) = \int_0^{\infty} e^{-st} f(t) dt$

$$\begin{aligned}
L[e^{at}f(t)] &= \int_0^{\infty} e^{-st} e^{at} f(t) dt \\
&= \int_0^{\infty} e^{-(s-a)t} f(t) dt \\
&= \varphi(s-a)
\end{aligned}$$

$$L[e^{-at}f(t)] = \int_0^{\infty} e^{-st} e^{-at} f(t) dt$$

$$\begin{aligned}
 &= \int_0^{\infty} e^{-(s+a)t} f(t) dt \\
 &= \int_0^{\infty} e^{-(s+a)t} f(t) dt \\
 &= \varphi(s+a)
 \end{aligned}$$

III. PROBLEMS BASED ON FIRST SHIFTING THEOREM AND SECOND SHIFTING THEOREM

Example Find $L[t^n e^{-at}]$

$$\begin{aligned}
 \text{Solution : } L[t^n e^{-at}] &= [L(t^n)]_{s \rightarrow (s+a)} \\
 &= \left[\frac{n!}{s^{n+1}} \right]_{s \rightarrow (s+a)} \\
 &= \frac{n!}{(s+a)^{n+1}}
 \end{aligned}$$

Example Find $L[e^{-at} \cos bt]$

$$\begin{aligned}
 \text{Solution : } L[e^{-at} \cos bt] &= [L[\cos bt]]_{s \rightarrow (s+a)} \\
 &= \left[\frac{s}{s^2 + b^2} \right]_{s \rightarrow (s+a)} \\
 &= \frac{s+a}{(s+a)^2 + b^2}
 \end{aligned}$$

Example Find $L[e^{at} \sinh bt]$

$$\begin{aligned}
 \text{Solution : } L[e^{at} \sinh bt] &= [L[\sinh bt]]_{s \rightarrow (s-a)} \\
 &= \left[\frac{b}{s^2 - b^2} \right]_{s \rightarrow (s-a)} = \frac{b}{(s-a)^2 - b^2}
 \end{aligned}$$

Example Find $L[e^t t^{-1/2}]$

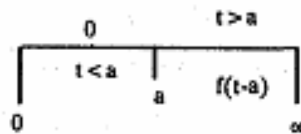
$$\begin{aligned}
 \text{Solution : } L[e^t t^{-1/2}] &= [L[t^{-1/2}]]_{s \rightarrow (s-1)} \\
 &= \left[\frac{\Gamma(-1/2+1)}{s^{-1/2+1}} \right]_{s \rightarrow (s-1)} = \left[\frac{\Gamma_{1/2}}{s^{1/2}} \right]_{s \rightarrow (s-1)} \\
 &= \left[\frac{\sqrt{\pi}}{\sqrt{s}} \right]_{s \rightarrow (s-1)} = \left[\sqrt{\frac{\pi}{s}} \right]_{s \rightarrow (s-1)} \\
 &= \sqrt{\frac{\pi}{s-1}}
 \end{aligned}$$

Result 14. Second shifting theorem.

- If $L[f(t)] = \varphi(s)$ and $G(t) = \begin{cases} f(t-a), & t > a \\ 0, & t < a \end{cases}$

then $L[G(t)] = e^{-as} \varphi(s)$

Proof : $L[G(t)] = \int_0^{\infty} e^{-st} G(t) dt$



$$= \int_0^a e^{-st} 0 dt + \int_a^{\infty} e^{-st} f(t-a) dt$$

$$= \int_a^{\infty} e^{-st} f(t-a) dt$$

Put $t-a = u$ $t \rightarrow a \Rightarrow u \rightarrow 0$
 $dt = du$ $t \rightarrow \infty \Rightarrow u \rightarrow \infty$

$$= \int_0^{\infty} e^{-s(u+a)} f(u) du$$

$$= e^{-sa} \int_0^{\infty} e^{-su} f(u) du$$

$$= e^{-sa} \int_0^{\infty} e^{-su} f(u) du \quad [u \text{ is a dummy variable}]$$

$$= e^{-sa} L[f(t)]$$

$$= e^{-sa} \varphi(s)$$

Result : 15. If $L[F(t)] = \varphi(s)$ and $C > 0$ then

$$L[F(t-c) H(t-c)] = e^{-cs} \varphi(s) \text{ where } H(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$$

Proof : $L[f(t)] = \int_0^{\infty} e^{-st} f(t) dt$

$$L[F(t-c) H(t-c)] = \int_0^{\infty} e^{-st} F(t-c) H(t-c) dt$$

DERIVATIVES AND INTEGRALS OF TRANSFORMS

TRANSFORMS OF DERIVATIVES AND INTEGRALS

Result : 17. Transforms of Derivatives

If $L[f(t)] = \varphi(s)$ then $L[tf(t)] = -\frac{d}{ds}\varphi(s) = -\varphi'(s)$

Proof : $\varphi(s) = L[f(t)]$

$$\frac{d}{ds}\varphi(s) = \frac{d}{ds}L[f(t)]$$

$$\begin{aligned}\varphi'(s) &= \frac{d}{ds} \left[\int_0^{\infty} e^{-st} f(t) dt \right] = \int_0^{\infty} \frac{\partial}{\partial s} (e^{-st}) f(t) dt \\ &= \int_0^{\infty} e^{-st} (-t) f(t) dt = - \int_0^{\infty} e^{-st} t f(t) dt \\ &= -L[tf(t)]\end{aligned}$$

$$\begin{aligned}\text{Put } t - c &= u & t \rightarrow 0 &\Rightarrow u \rightarrow -c \\ dt &= du & t \rightarrow \infty &\Rightarrow u \rightarrow \infty\end{aligned}$$

$$= \int_{-c}^{\infty} e^{-s(u+c)} F(u) H(u) du$$

$$= e^{-sc} \int_{-c}^{\infty} e^{-su} F(u) H(u) du$$

$$= e^{-sc} \left[\int_{-c}^0 e^{-su} F(u) 0 du + \int_0^{\infty} e^{-su} F(u) du \right]$$

$$= e^{-sc} \int_0^{\infty} e^{-su} F(u) du$$

$$= e^{-sc} \int_0^{\infty} e^{-st} F(t) dt \quad [\because u \text{ is a dummy variable}]$$

$$= e^{-sc} L[F(t)] = e^{-sc} \varphi(s)$$

$$L[tf(t)] = -\varphi'(s)$$

Corollary :- If $L[f(t)] = \varphi(s)$ then $L[t^n f(t)] = (-1)^n \varphi^n(s)$.

Proof : W.K.T. $L[tf(t)] = -\varphi'(s)$

$$\begin{aligned} L[t^2 f(t)] &= L[t \cdot tf(t)] \\ &= -\frac{d}{ds} L[tf(t)] \\ &= -\frac{d}{ds} \left[-\frac{d}{ds} Lf(t) \right] \\ &= (-1)^2 \frac{d^2}{ds^2} [Lf(t)] \\ &= (-1)^2 \frac{d^2}{ds^2} \varphi(s) \end{aligned}$$

.....

$$L[t^n f(t)] = (-1)^n \frac{d^n}{ds^n} \varphi(s) = (-1)^n \varphi^n(s)$$

PROBLEMS BASED ON TRANSFORMS OF DERIVATIVES

Example 1. Find $L[t \sin 2t]$

Solution : W.K.T. $L[t^n f(t)] = (-1)^n \varphi^n(s)$

$$\begin{aligned} L(t \sin 2t) &= -\frac{d}{ds} [L(\sin 2t)] = -\frac{d}{ds} \left[\frac{2}{s^2 + 4} \right] \\ &= -\left[\frac{-4s}{(s^2 + 4)^2} \right] = \frac{4s}{(s^2 + 4)^2} \end{aligned}$$

Example 2. Find $L[t^2 e^{-3t}]$

Solution : W.K.T $L[t^n f(t)] = (-1)^n \frac{d^n}{ds^n} [\varphi(s)]$

$$\begin{aligned} L[t^2 e^{-3t}] &= (-1)^2 \frac{d^2}{ds^2} L[e^{-3t}] = \frac{d^2}{ds^2} \left[\frac{1}{s + 3} \right] \\ &= \frac{d}{ds} \left[\frac{-1}{(s + 3)^2} \right] = \frac{2}{(s + 3)^3} \end{aligned}$$

Example 3. Find $L [te^{-2t} \sin t]$

$$\begin{aligned}\text{Solution : } L[te^{-2t} \sin t] &= -\frac{d}{ds} [L(e^{-2t} \sin t)] \\ &= -\frac{d}{ds} \left[L[\sin t] \right]_{s \rightarrow (s+2)} = -\frac{d}{ds} \left[\left[\frac{1}{s^2 + 1} \right]_{s \rightarrow (s+2)} \right] \\ &= -\frac{d}{ds} \left[\frac{1}{(s+2)^2 + 1} \right] = \frac{2(s+2)}{[(s+2)^2 + 1]^2}\end{aligned}$$

Example Find $L [t \sin 3t \cos 2t]$

$$\begin{aligned}\text{Solution : } L[t \sin 3t \cos 2t] &= -\frac{d}{ds} [L(\sin 3t \cos 2t)] \\ &= -\frac{d}{ds} \left[\frac{1}{2} [L(\sin 5t) + L(\sin t)] \right] = -\frac{1}{2} \frac{d}{ds} \left[\frac{5}{s^2 + 25} + \frac{1}{s^2 + 1} \right] \\ &= \frac{5s}{(s^2 + 25)^2} + \frac{s}{(s^2 + 1)^2}\end{aligned}$$

Example 5. Given that $L[\sin \sqrt{t}] = \frac{1}{2s} \sqrt{\frac{\pi}{s}} e^{-1/4s}$ find L.T. of $\frac{1}{\sqrt{t}} \cos \sqrt{t}$

Solution : Let $f(t) = \sin \sqrt{t}$

$$f'(t) = \frac{1}{2\sqrt{t}} \cos \sqrt{t}$$

$$L[f'(t)] = s L[f(t)] - f(0)$$

$$\begin{aligned}L\left[\frac{1}{2\sqrt{t}} \cos \sqrt{t}\right] &= L[f'(t)] \\ &= s \frac{1}{2s} \sqrt{\frac{\pi}{s}} e^{-1/4s} - 0 \quad [\because f(0) = 0] \\ &= \frac{1}{2} \sqrt{\frac{\pi}{s}} e^{-1/4s}\end{aligned}$$

$$\begin{aligned}\frac{1}{2} L\left[\frac{1}{\sqrt{t}} \cos \sqrt{t}\right] &= \frac{1}{2} \sqrt{\frac{\pi}{s}} e^{-1/4s} \\ L\left[\frac{1}{\sqrt{t}} \cos \sqrt{t}\right] &= \sqrt{\frac{\pi}{s}} e^{-1/4s}\end{aligned}$$

Example 6 show that $\int_0^{\infty} e^{-t} t \cos t \, dt = 0$

Solution

$$\begin{aligned} \int_0^{\infty} t \cos t \, dt &= \left[L[t \cos t] \right]_{s=1} \quad \text{Final Value Theorem} \\ &= \left[-\frac{d}{ds} L(\cos t) \right]_{s=1} = \left[-\frac{d}{ds} \left(\frac{s}{s^2+1} \right) \right]_{s=1} = \left[-\frac{(s^2+1)(1) - s(2s)}{(s^2+1)^2} \right]_{s=1} \\ &= \left[-\frac{s^2+1-2s^2}{(s^2+1)^2} \right]_{s=1} = \left[-\frac{1-s^2}{(s^2+1)^2} \right]_{s=1} = [-(0)] = 0 \end{aligned}$$

Example 7 Find $L[te^{-t} \cosh t]$

Solution :

$$\begin{aligned} L[te^{-t} \cosh t] &= -\frac{d}{ds} L[e^{-t} \cosh t] \\ &= -\frac{d}{ds} \left[\frac{s+1}{(s+1)^2-1} \right] = -\frac{[(s+1)^2-1] - (s+1)2(s+1)}{[(s+1)^2-1]^2} \\ &= -\frac{(s+1)^2-1-2(s+1)^2}{[(s+1)^2-1]^2} = \frac{(s+1)^2+1}{(s^2+2s)^2} = \frac{s^2+2s+2}{s^4+4s^2+4s^3} \end{aligned}$$

Result 18. Integrals of transform

If $L[f(t)] = \varphi(s)$ and $\frac{1}{t}f(t)$ has a limit as $t \rightarrow 0$ then

$$L\left[\frac{1}{t}f(t)\right] = \int_s^{\infty} \varphi(s) \, ds$$

Proof : $\varphi(s) = L[f(t)]$

$$\begin{aligned} \int_s^{\infty} \varphi(s) \, ds &= \int_s^{\infty} L[f(t)] \, ds \\ &= \int_s^{\infty} \int_0^{\infty} e^{-st} f(t) \, dt \, ds = \int_0^{\infty} \int_s^{\infty} e^{-st} f(t) \, ds \, dt \end{aligned}$$

[since s and t are independent variables and hence the order of integration in the double integral can be interchanged]

$$\begin{aligned} &= \int_0^{\infty} f(t) \left[\int_s^{\infty} e^{-st} \, ds \right] dt = \int_0^{\infty} f(t) \left[\frac{e^{-st}}{-t} \right]_s^{\infty} dt \\ &= \int_0^{\infty} f(t) \left[0 + \frac{e^{-st}}{t} \right] dt = \int_0^{\infty} f(t) \frac{e^{-st}}{t} dt \\ &= \int_0^{\infty} e^{-st} \frac{f(t)}{t} dt = L\left[\frac{1}{t}f(t)\right] \end{aligned}$$

$$\text{i.e., } L \left[\frac{1}{t} f(t) \right] = \int_s^\infty \varphi(s) ds$$

PROBLEMS BASED ON INTEGRALS OF TRANSFORM

Example 8 Find $L \left[\frac{1 - e^t}{t} \right]$

Solution : $L \left[\frac{1}{t} f(t) \right] = \int_s^\infty \varphi(s) ds = \int_s^\infty L[f(t)] ds$

$$\begin{aligned} L \left[\frac{1 - e^t}{t} \right] &= \int_s^\infty L[1 - e^t] ds = \int_s^\infty \left[\frac{1}{s} - \frac{1}{s-1} \right] ds \\ &= \left[\log s - \log(s-1) \right]_s^\infty = \left[\log \frac{s}{s-1} \right]_s^\infty \\ &= \left[\log \frac{s}{s(1-1/s)} \right]_s^\infty = \left[\log \frac{1}{1-1/s} \right]_s^\infty \\ &= 0 - \log \frac{s}{s-1} = \log \left(\frac{s-1}{s} \right) \end{aligned}$$

Example 9 Find $L \left[\frac{\sin at}{t} \right]$ [A.U., March 1996]

Solution : $L \left[\frac{1}{t} f(t) \right] = \int_s^\infty L[f(t)] ds$

$$\begin{aligned} L \left[\frac{\sin at}{t} \right] &= \int_s^\infty L[\sin at] ds = \int_s^\infty \frac{a}{s^2 + a^2} ds \\ &= a \left[\frac{1}{a} \tan^{-1} \left(\frac{s}{a} \right) \right]_s^\infty = \left[\tan^{-1} \frac{s}{a} \right]_s^\infty \\ &= \frac{\pi}{2} - \tan^{-1} \left(\frac{s}{a} \right) = \cot^{-1} \left[\frac{s}{a} \right] = \tan^{-1} \left[\frac{a}{s} \right] \end{aligned}$$

Note : $\cot^{-1} \left(\frac{s}{a} \right) = \frac{\pi}{2} - \tan^{-1} \left(\frac{s}{a} \right)$

$$\begin{aligned} \frac{s}{a} &= \cot \left[\frac{\pi}{2} - \tan^{-1} \left(\frac{s}{a} \right) \right] \\ &= \tan \left[\tan^{-1} \left(\frac{s}{a} \right) \right] = \frac{s}{a} \end{aligned}$$

INITIAL AND FINAL VALUE THEOREMS

◆ INITIAL VALUE THEOREM

If $L[f(t)] = F(s)$, then $\lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} s F(s)$

Proof : W.K.T.

$$\begin{aligned} L[f'(t)] &= s L[f(t)] - f(0) \\ &= s F(s) - f(0) \\ s F(s) - f(0) &= L[f'(t)] \\ &= \int_0^{\infty} e^{-st} f'(t) dt \end{aligned}$$

$$\lim_{s \rightarrow \infty} [s F(s) - f(0)] = \lim_{s \rightarrow \infty} \int_0^{\infty} e^{-st} f'(t) dt$$

$$\lim_{s \rightarrow \infty} s F(s) - f(0) = 0 \quad [\because e^{-\infty} = 0]$$

$$\text{i.e., } \lim_{s \rightarrow \infty} s F(s) = f(0) = \lim_{t \rightarrow 0} f(t)$$

$$\text{Hence } \lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} s F(s)$$

◆ FINAL VALUE THEOREM

If $L[f(t)] = F(s)$, then $\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s F(s)$

Proof : W.K.T. $L[f'(t)] = s L[f(t)] - f(0)$

$$s L[f(t)] - f(0) = L[f'(t)] = \int_0^{\infty} e^{-st} f'(t) dt$$

$$\begin{aligned} \lim_{s \rightarrow 0} [s L[f(t)] - f(0)] &= \lim_{s \rightarrow 0} \int_0^{\infty} e^{-st} f'(t) dt \\ &= \int_0^{\infty} f'(t) dt = \int_0^{\infty} d[f(t)] = f(t) \Big|_0^{\infty} \end{aligned}$$

$$\lim_{s \rightarrow 0} s F(s) - f(0) = f(\infty) - f(0)$$

$$\lim_{s \rightarrow 0} s F(s) = f(\infty) = \lim_{t \rightarrow \infty} f(t)$$

$$\text{Hence } \lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s F(s)$$

PROBLEMS BASED ON INITIAL VALUE AND FINAL VALUE THEOREM

Example 5.4.1. If $L[f(t)] = \frac{1}{s(s+a)}$, find $\lim_{t \rightarrow \infty} f(t)$ and $\lim_{t \rightarrow 0} f(t)$

Solution : $\lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} s \cdot F(s)$

$$= \lim_{s \rightarrow \infty} s \frac{1}{s(s+a)} = \lim_{s \rightarrow \infty} \frac{1}{s+a} = \frac{1}{\infty} = 0$$

$$\begin{aligned} \lim_{t \rightarrow \infty} f(t) &= \lim_{s \rightarrow 0} s F(s) = \lim_{s \rightarrow 0} s \frac{1}{s(s+a)} \\ &= \lim_{s \rightarrow 0} \frac{1}{s+a} = \frac{1}{a} \end{aligned}$$

Example 2. Verify the initial and final value theorem for the function

$$f(t) = 1 + e^{-t}(\sin t + \cos t)$$

Solution : Initial value theorem states that

$$\lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} s F(s)$$

$$L[f(t)] = F(s) = \frac{1}{s} + L[\sin t + \cos t]_{s \rightarrow s+1}$$

$$= \frac{1}{s} + \frac{1}{(s+1)^2 + 1} + \frac{s+1}{(s+1)^2 + 1}$$

$$= \frac{1}{s} + \frac{s+2}{(s+1)^2 + 1}$$

$$\text{L.H.S} = \lim_{t \rightarrow 0} f(t) = 1 + 1 = 2$$

$$\text{R.H.S} = \lim_{s \rightarrow \infty} s \left[\frac{1}{s} + \frac{s+2}{(s+1)^2 + 1} \right]$$

$$= \lim_{s \rightarrow \infty} \left[1 + \frac{s(s+2)}{(s+1)^2 + 1} \right] = \lim_{s \rightarrow \infty} \left[1 + \frac{s^2 \left(1 + \frac{2}{s}\right)}{s^2 \left[1 + \frac{2}{s} + \frac{2}{s^2}\right]} \right]$$

$$= \lim_{s \rightarrow \infty} \left[1 + \frac{1 + \frac{2}{s}}{1 + \frac{2}{s} + \frac{2}{s^2}} \right] = 1 + 1 = 2$$

$$\text{L.H.S.} = \text{R.H.S.}$$

Initial value theorem verified.

$$\begin{aligned}
 \lim_{t \rightarrow \infty} f(t) &= \lim_{s \rightarrow 0} s F(s) \\
 \text{L.H.S.} &= \lim_{t \rightarrow \infty} [1 + e^{-t} (\sin t + \cos t)] \\
 &= 1 + 0 = 1 \\
 \text{R.H.S.} &= \lim_{s \rightarrow 0} \left[1 + \frac{s(s+2)}{(s+1)^2 + 1} \right] \\
 &= 1 + 0 = 1 \\
 \text{L.H.S.} &= \text{R.H.S.}
 \end{aligned}$$

Final value theorem verified.

Example 3. Verify the initial and final value theorems for $f(t) = 3e^{-2t}$

Solution : $f(t) = 3e^{-2t}$

$$F(s) = L[f(t)] = L[3e^{-2t}] = \frac{3}{s+2}$$

Initial value theorem : $\lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} s F(s)$

$$\text{L.H.S.} = \lim_{t \rightarrow 0} f(t) = \lim_{t \rightarrow 0} 3e^{-2t} = 3$$

$$\text{R.H.S.} = \lim_{s \rightarrow \infty} s F(s) = \lim_{s \rightarrow \infty} s \left(\frac{3}{s+2} \right) = \lim_{s \rightarrow \infty} \frac{3s}{s+2}$$

$$= \lim_{s \rightarrow \infty} \frac{3s}{s \left(1 + \frac{2}{s} \right)}$$

$$= \lim_{s \rightarrow \infty} \frac{3}{1 + \left(\frac{2}{s} \right)} = 3$$

$$\text{L.H.S.} = \text{R.H.S.}$$

Hence Initial value theorem verified.

Final value theorem $\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} s F(s)$

Mathematics II

$$\text{L.H.S.} = \lim_{t \rightarrow \infty} f(t) = \lim_{t \rightarrow \infty} 3e^{-2t} = 0 \quad [\because e^{-\infty} = 0]$$

$$\text{R.H.S.} = \lim_{s \rightarrow 0} s F(s) = \lim_{s \rightarrow 0} s \left(\frac{3}{s+2} \right) = 0$$

$$\text{L.H.S.} = \text{R.H.S.}$$

Hence Final value theorem verified.

TRANSFORMS OF UNIT STEP FUNCTION AND IMPULSE FUNCTION

◆ UNIT STEP FUNCTION (OR) HEAVISIDE'S UNIT STEP FUNCTION

PROBLEMS BASED ON UNIT STEP FUNCTION

Example 1. Define the unit step function.

Solution :

The unit step function, also called Heaviside's unit function is defined as

$$U(t-a) = \begin{cases} 0 & \text{for } t < a \\ 1 & \text{for } t > a \end{cases}$$

This is the unit step functions at $t = a$

It can also be denoted by $H(t-a)$.

Example 2. Give the L.T. of the unit step function. [M.U. Oct., 96]

Solution :

The L.T. of the unit step function is given by

$$\begin{aligned} L[U(t-a)] &= \int_0^{\infty} e^{-st} U(t-a) dt \\ &= \int_0^a e^{-st} 0 dt + \int_a^{\infty} e^{-st} (1) dt \\ &= \int_a^{\infty} e^{-st} dt = \left[\frac{e^{-st}}{-s} \right]_a^{\infty} \\ &= 0 - \left(\frac{e^{-sa}}{-s} \right) = \frac{e^{-as}}{s} \end{aligned}$$

TRANSFORM OF PERIODIC FUNCTIONS

Define periodic function and state its Laplace transform formula.

◆ Def. Periodic

A function $f(x)$ is said to be "periodic" if and only if $f(x+p) = f(x)$ is true for some value of p and every value of x . The smallest positive value of p for which this equation is true for every value of x will be called the period of the function.

The Laplace Transformation of a periodic function $f(t)$ with period p given by $\frac{1}{1 - e^{-ps}} \int_0^p e^{-st} f(t) dt$

$$\begin{aligned} \text{Proof : } L[f(t)] &= \int_0^{\infty} e^{-st} f(t) dt \\ &= \int_0^p e^{-st} f(t) dt + \int_p^{\infty} e^{-st} f(t) dt \end{aligned}$$

Put $t = u + p$ in the second integral

$$\text{i.e., } u = t - p \quad t \rightarrow p \Rightarrow u \rightarrow 0$$

$$\text{i.e., } du = dt \quad t \rightarrow \infty \Rightarrow u \rightarrow \infty$$

$$\begin{aligned} &= \int_0^p e^{-st} f(t) dt + \int_0^{\infty} e^{-(u+p)s} f(u+p) du \\ &= \int_0^p e^{-st} f(t) dt + e^{-sp} \int_0^{\infty} e^{-su} f(u) du \quad [\because f(u+p) = f(u)] \\ &= \int_0^p e^{-st} f(t) dt + e^{-sp} \int_0^{\infty} e^{-st} f(t) dt \quad [\because u \text{ is a dummy variable}] \end{aligned}$$

$$L[f(t)] = \int_0^p e^{-st} f(t) dt + e^{-sp} L[f(t)]$$

$$[1 - e^{-sp}] L[f(t)] = \int_0^p e^{-st} f(t) dt$$

$$L[f(t)] = \frac{1}{1 - e^{-sp}} \int_0^p e^{-st} f(t) dt$$

Example 1 Find the Laplace transform of the Half wave rectifier function

$$f(t) = \begin{cases} \sin \omega t, & 0 < t < \frac{\pi}{\omega} \\ 0, & \frac{\pi}{\omega} < t < \frac{2\pi}{\omega} \end{cases}$$

Solution : This function is a periodic function with period $\frac{2\pi}{\omega}$ in the interval $\left(0, \frac{2\pi}{\omega}\right)$

$$\begin{aligned} \mathcal{L}[f(t)] &= \frac{1}{1 - e^{-\frac{2\pi s}{\omega}}} \int_0^{\frac{2\pi}{\omega}} e^{-st} f(t) dt \\ &= \frac{1}{1 - e^{-\frac{2\pi s}{\omega}}} \left[\int_0^{\pi/\omega} e^{-st} \sin \omega t dt + 0 \right] \\ &= \frac{1}{1 - e^{-\frac{2\pi s}{\omega}}} \left[\frac{e^{-st}}{s^2 + \omega^2} [-s \sin \omega t - \omega \cos \omega t] \right]_0^{\pi/\omega} \\ &= \frac{1}{1 - e^{-\frac{2\pi s}{\omega}}} \left[\frac{e^{-s\pi/\omega} \omega + \omega}{s^2 + \omega^2} \right] \\ &= \frac{\omega [1 + e^{-\frac{s\pi}{\omega}}]}{[1 - e^{-s\pi/\omega}] [1 + e^{-s\pi/\omega}] (s^2 + \omega^2)} \\ &= \frac{\omega}{(s^2 + \omega^2) (1 - e^{-s\pi/\omega})} \end{aligned}$$

Example 2 Find the Laplace Transform of

$$f(t) = \begin{cases} 1, & 0 < t < a \\ 2a-t, & a < t < 2a \end{cases} \text{ with } f(t+2a) = f(t)$$

$$\begin{aligned} \text{Solution : } L[f(t)] &= \frac{1}{1-e^{-2as}} \int_0^{2a} e^{-st} f(t) dt \\ &= \frac{1}{1-e^{-2as}} \left[\int_0^a e^{-st} t dt + \int_a^{2a} e^{-st} (2a-t) dt \right] \\ &= \frac{1}{1-e^{-2as}} \left[\left[t \left(\frac{e^{-st}}{-s} \right) - (-1) \left(\frac{e^{-st}}{s^2} \right) \right]_0^a + \left[(2a-t) \left(\frac{e^{-st}}{-s} \right) - (-1) \left(\frac{e^{-st}}{s^2} \right) \right]_a^{2a} \right] \\ &= \frac{1}{1-e^{-2as}} \left[\left[-t \frac{e^{-st}}{s} - \frac{e^{-st}}{s^2} \right]_0^a + \left[-(2a-t) \frac{e^{-st}}{s} + \frac{e^{-st}}{s^2} \right]_a^{2a} \right] \\ &= \frac{1}{1-e^{-2as}} \left[\left(-a \frac{e^{-as}}{s} - \frac{e^{-as}}{s^2} \right) - \left(-\frac{1}{s^2} \right) + \left[\left(\frac{e^{-2as}}{s^2} \right) - \left(-a \frac{e^{-as}}{s} + \frac{e^{-as}}{s^2} \right) \right] \right] \\ &= \frac{1}{1-e^{-2as}} \left[-\frac{ae^{-as}}{s} - \frac{e^{-as}}{s^2} + \frac{1}{s^2} + \frac{e^{-2as}}{s^2} + a \frac{e^{-as}}{s} - \frac{e^{-as}}{s^2} \right] \\ &= \frac{1}{1-e^{-2as}} \left[\frac{1+e^{-2as}-2e^{-as}}{s^2} \right] \\ &= \frac{[1-e^{-as}]^2}{s^2(1-e^{-as})(1+e^{-as})} = \frac{1-e^{-as}}{s^2(1+e^{-as})} = \frac{1}{s^2} \tanh \left[\frac{as}{2} \right] \end{aligned}$$

